

A top-down view of a grey desk with various items: a stethoscope, a laptop with a tablet on top, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

**Pharm***Achieve*

# FIBROMYALGIA

Qualifying Exam Preparatory Course

# COMPETENCIES COVERED IN THIS PRESENTATION...

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**Providing Care:  
Clinical Care**

**Providing Care:  
Distribution**

**Knowledge  
& Expertise**

**Communication  
& Collaboration**

**Leadership &  
Stewardship**

**Professionalism**

# LEGEND

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**BID** Twice Daily

**CBC** Complete Blood Count

**CBT** Cognitive Behavioural Therapy

**CK** Creatine Kinase

**CRP** C-Reactive Protein

**ESR** Erythrocyte Sedimentation Rate

**NSAID** Nonsteroidal Anti-Inflammatory Drug

**QHS** Every Night At Bedtime

**SNRI** Serotonin Norepinephrine Reuptake Inhibitor

**SSRI** Selective Serotonin Reuptake Inhibitor

**TCA** Tricyclic Antidepressant

**TSH** Thyroid Stimulating Hormone

# PATHOPHYSIOLOGY

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A **chronic condition** characterized by:

- Widespread **musculoskeletal pain**
- **Disordered sleep & fatigue**
- **Cognitive dysfunction**
- **Mood changes** (may be due to maladaptive neuroplasticity)

**Direct cause is unknown**

- Involves immune, endocrine and nervous system

## Possible triggers

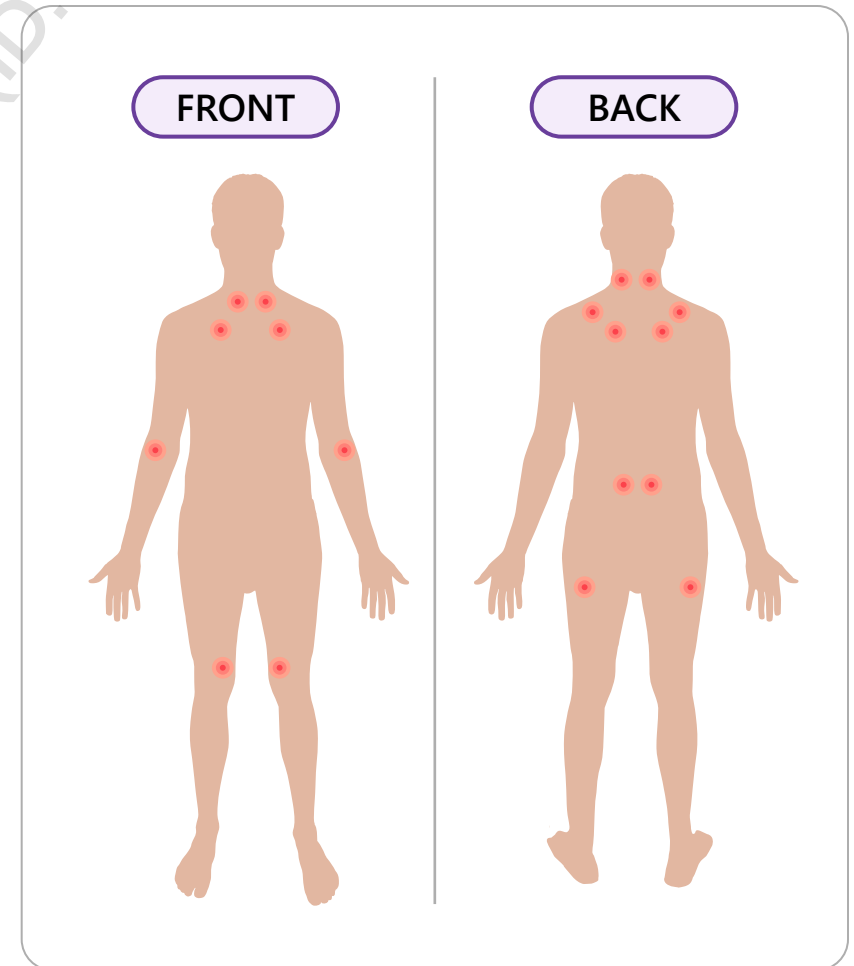
- Trauma
- Neurochemical imbalances
- Infections (e.g. Lyme disease, viral hepatitis)
- Psychological stressors
- Adverse life events
- Family history

# CLINICAL PRESENTATION

- Widespread musculoskeletal pain
- Fatigue and sleep disturbances
- Distressed mood: anxiety, depression
- Cognitive disturbances: memory issues, difficulty concentrating, confusion ("brain fog" or "fibro fog")



No consistent pattern of symptomatology



# RISK FACTORS

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Gender  
(females > males)

Age  
(incidence increases with age)

Most prevalent in **female**  
patients in their **30s to 50s**

Family history

Rheumatic disease  
(e.g. rheumatoid arthritis or  
lupus)

# DIAGNOSIS

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FYI

- Suspect in patients with widespread chronic musculoskeletal pain for at least 3 months
- **Diagnosis based on clinical construct**
  - Thorough history and physical examination to rule out other conditions
    - Simple blood testing (e.g. CBC, ESR, CRP, CK, TSH) to rule out other conditions
    - Consider referral to subspecialist
  - No confirmatory laboratory test
- Primary care physician to establish diagnosis – no need for specialist
  - Specialist may be required for consultation re: sleep, but continued care by a specialist not recommended
  - Repeated investigations after diagnosis should be avoided unless driven by new onset of symptoms

# MEDICAL/DRUG RELATED CAUSES

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- Infections (bacterial/viral)
- Rheumatoid arthritis
- Lupus
- Hypothyroidism
- Hyperparathyroidism
- Polymyalgia rheumatica
- Giant-cell arteritis
- Spondyloarthritis
- Metastases
- Vitamin D deficiency
- Systemic vasculitis
- Irritable bowel syndrome
- Depression
- Statins
- Aromatase inhibitors
- Bisphosphonates



# GOALS OF THERAPY

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- ✓ Alleviate pain
- ✓ Resolve sleeping problems and psychological distress
- ✓ Reduce fatigue
- ✓ Improve daily functioning and quality of life
- ✓ Provide patient education and self-management strategies

# NON-PHARMACOLOGICAL MEASURES

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- Address comorbidities and drug-related causes
- Exercise programs: graded aerobic exercises, pool exercises, walking programs, strength training, flexibility
- Sleep hygiene
- Stress management: meditative movement therapies (e.g. yoga)
- Psychological therapy: CBT, mindfulness-based stress reduction (MBSR), meditation, relaxation
  - Cognitive Behavioural Therapy (CBT)
    - Multimodal approach that alters patterns of negative thoughts and attitudes by fostering more adaptive thoughts and attitudes
    - Addresses pain and improves functioning and self-management skills
    - Significantly reduces pain severity
- Non-drug pain reduction strategies: massages
- Complementary and alternative medicines: acupuncture, chiropractic therapy, supplements (insufficient evidence)

Non-drug measures are **first-line** as strong data to support the use of drug therapy is not available

# PHARMACOLOGICAL MEASURES

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## Antidepressants

- TCAs, SSRIs, SNRIs, quetiapine, trazodone
- **Duloxetine** is Health Canada approved

## GABA derivatives

- **Pregabalin** (Health Canada approved), gabapentin

## Other

- Cyclobenzaprine, acetaminophen, opioids, lidocaine injection, cannabinoids, zopiclone, human growth hormone

# ANTIDEPRESSANTS

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- TCAs (e.g. amitriptyline)
  - Low doses administered QHS
  - Anticholinergic effects (e.g. dry mouth, constipation, urinary retention)
  - Short-term efficacy in improving sleep and pain
- Cyclobenzaprine (structurally similar to TCAs)
  - Administered QHS
  - Moderate benefits in sleep, fatigue, and depression
- SSRIs/SNRIs
  - Better tolerated than TCAs
  - Less effective in treating pain associated with fibromyalgia
  - Duloxetine (Health-Canada approved)- first line in comorbid fibromyalgia and depression
    - Shown to reduce pain, improve mood, and is well-tolerated
    - Not shown to improve quality of life, fatigue, or sleep disturbance

# GABA DERIVATIVES

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- Pregabalin
  - Health Canada approved for treating fibromyalgia
  - Shown to reduce pain, improve sleep, and quality of life
  - Moderate doses of pregabalin are required to show clinical efficacy
    - Titrate slowly and initiate therapy at bedtime to minimize side effects such as drowsiness
- Gabapentin
  - Less evidence than pregabalin
  - Shown to improve sleep and pain scores
  - Significantly elevated doses may be required

# OTHER PHARMACOLOGICAL MEASURES

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## Analgesics (e.g. acetaminophen)

- No direct evidence in treating fibromyalgia pain
- Unlikely to be harmful if taken at usual dose

## Opioids

- Tramadol (with or without acetaminophen)
  - Only opioid that reduces pain and improves quality of life in patients
  - Reduces serotonin and norepinephrine reuptake
- No good evidence is available that other opioids are beneficial. Some fibromyalgia guidelines are against using opioids for treatment
- Potential for opioid-induced hyperalgesia as patients with fibromyalgia may already be experiencing central sensitization (hyperalgesia)

# OTHER PHARMACOLOGICAL MEASURES

FYI

## Sedatives

- Can be used in patients having trouble sleeping
- Concerns of dependency and adverse effects
- **Zopiclone** can be used intermittently (insufficient evidence)
- **Trazodone** is an example of antidepressant with sedative properties (insufficient evidence)
- **Quetiapine** used off-label for sleep reduced stiffness and fatigue, but not pain

## Cannabinoids

- May be tried in a patient with fibromyalgia in the setting of sleep disturbances  
The College of Family Physicians of Canada **strongly advises against this**

## Human growth hormones

- Further research is required but a placebo-controlled trial showed improved symptoms
- May only help patients who have low levels of growth hormones
- Limited in use due to availability, expense, and safety concerns

## Local injections

- Lidocaine 1% with/without a depot form of corticosteroid to the myofascial trigger points can be useful in reducing pain and the perpetuation of central pain sensitization

# PREGNANCY AND BREASTFEEDING

## Pregnancy

- Non-pharmacological measures are strongly encouraged
- Acetaminophen can be used
- Limited evidence on the safety of pregabalin, gabapentin and amitriptyline
  - Assess benefits vs risks



## Breastfeeding

- Acetaminophen is compatible
- Not recommended: pregabalin, gabapentin, tramadol, amitriptyline





# MONITORING PARAMETERS

| Parameter                                                                        | Degree of Change                                  | Timeframe |
|----------------------------------------------------------------------------------|---------------------------------------------------|-----------|
| Fibromyalgia symptoms (e.g. chronic pain, fatigue, cognitive dysfunction, sleep) | Improve<br><br>Moderate pain reduction<br>(≥ 30%) | Ongoing   |
| Side effects of medications                                                      | Prevent/minimize                                  |           |



The development of a new symptom requires clinical evaluation to ensure that symptoms are not due to some other medical illness

# TAKEAWAY

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- Individualized tailored treatment approach → Symptom-based management
- Non-pharmacological measures are 1<sup>st</sup> line
  - Pharmacologic agents are most effective when combined with non-pharmacological therapies
  - Insufficient evidence to support complementary and alternative medicines
- Patients may require a combination drug therapy regimen in which the drugs have different methods of action
- Consider patient symptoms, comorbidities and side effects of drugs when selecting therapy
  - If the patient has comorbid depression, consider use of antidepressant such as duloxetine
  - If patient has trouble sleeping, consider use of GABA derivative (pregabalin), cyclobenzaprine or amitriptyline
- Patients with fibromyalgia may be more sensitive to drug side effects → start therapies at low doses and titrate up slowly
- Help patients recognize factors that cause exacerbations of their symptoms
- Physicians should encourage remaining in workforce – offer support

# CASE 1

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FM is a 53-year-old female who presents explaining she has not been feeling refreshed upon waking although she sleeps for 8 hours a night. She has also been feeling “foggy,” and has been feeling tenderness all over her body. Upon asking, she explains she has not really noticed a difference in mood. She has a past medical history of hypothyroidism and leg cramps. She takes levothyroxine 150 mcg PO daily and magnesium oxide 420 mg PO at night. She exercises once per week. She has no allergies to any medications. She is seeking your advice.

1. Which of the following is not considered an appropriate non-pharm to recommend to FM?
  - a) Empathy and acknowledgement of suffering
  - b) A multidisciplinary program of education and self-management
  - c) Unsupervised aerobic exercises, walking programs, and aquatic exercises
  - d) Ongoing CBT sessions if applicable



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2. Which of the following have been approved by Health Canada for fibromyalgia treatment?
- a) Gabapentin and Pregabalin
  - b) Pregabalin and Venlafaxine
  - c) Duloxetine and Venlafaxine
  - d) Pregabalin and Duloxetine



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3. Which of the following targets would be the best 1<sup>st</sup> line option for treating FM based on her symptoms?
- a) Zopiclone 7.5 mg PO QHS
  - b) Duloxetine 60 mg PO daily
  - c) Pregabalin 150 mg PO BID
  - d) Venlafaxine 75 mg PO daily



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4. Which of the following is considered the least appropriate option in treating sleep disturbances in fibromyalgia patients?
- a) Gabapentin
  - b) Oxycodone
  - c) Amitriptyline
  - d) Trazodone



# REFERENCES

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1. Fibromyalgia. In: RxTx. Ottawa, ON: Canadian Pharmacists Association. <https://myrxtx.ca/new/documents/CHAPTER/en/c0053>.
2. Fitzcharles M, Ste-Marie P, Goldenberg D, Pereira J. Management | Canadian Guidelines for the Diagnosis and Management of Fibromyalgia. *National Guidelines*. 2012. Available at: [http://fmguidelines.ca/?page\\_id=21](http://fmguidelines.ca/?page_id=21).
3. Clauw DJ. Fibromyalgia: an overview. *Am J Med* 2009;122(12 Suppl):S3-13.
4. Fitzcharles MA, Ste-Marie PA, Pereira JX et al. Fibromyalgia: evolving concepts over the past 2 decades. *CMAJ* 2013;185(13):E645-51.
5. Kodner C. Common questions about the diagnosis and management of fibromyalgia. *Am Fam Physician* 2015;91(7):472-8.
6. Macfarlane GJ, Kronisch C, Dean LE et al. EULAR revised recommendations for the management of fibromyalgia. *Ann Rheum Dis* 2017;76(2):318-28.
7. Wolfe F, Clauw DJ, Fitzcharles M et al. 2016 revisions to the 2010/2011 fibromyalgia diagnostic criteria. *Semin Arthritis Rheum* 2016;46(3):319-329.
8. Jurado-Priego LN, Cueto-Ureña C, Ramírez-Expósito MJ, Martínez-Martos JM. Fibromyalgia: A Review of the Pathophysiological Mechanisms and Multidisciplinary Treatment Strategies. *Biomedicines*. 2024;12(7):1543. Published 2024 Jul 11. doi:10.3390/biomedicines12071543.
9. Moore A, Bidonde J, Fisher E, Häuser W, Bell RF, Perrot S, Makri S, Straube S. Effectiveness of pharmacological therapies for fibromyalgia syndrome in adults: an overview of Cochrane Reviews. *Rheumatology (Oxford)*. 2025 May 1;64(5):2385-2394. doi: 10.1093/rheumatology/keae707. PMID: 39705187; PMCID: PMC12048062.

# CHANGE LOG

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## August 2025

- Slide 7: Added 'FYI' to slide

## November 2025

- Minor formatting changes
- Slide 17: Minor edit to monitoring parameters
- Slide 18: Consolidated Takeaways into one slide
- Updated references

## April 2026

- Slide 2: Updated NAPRA competencies
- Slide 6: Changed women and men to females and males
- Slide 15: Added FYI as this material is less likely to be relevant for exam
- Slide 16: Simplified information